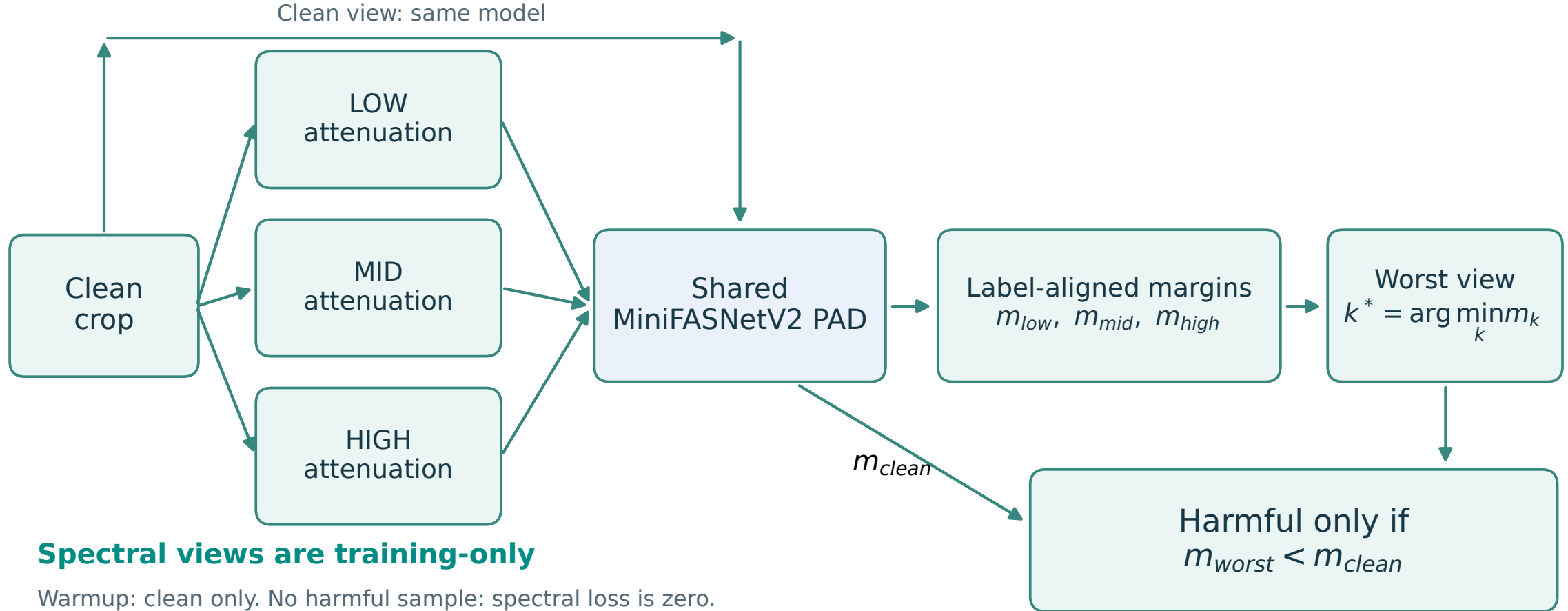


MAST-PAD: margin-aware spectral supervision

Training: choose the spectral view that most degrades the label-aligned margin



Spectral views are training-only

Warmup: clean only. No harmful sample: spectral loss is zero.

$$L = 0.75 CE_{clean} + 0.25 CE_{harmful} + L_{aux}$$

Inference: crop1.5 $\rightarrow$ 80 $\times$ 80 $\rightarrow$ MiniFASNetV2 $\rightarrow$ Real / Attack